Great job finishing Unit 6! ⍰ Here's a **comprehensive set of 30 questions** based on everything you've studied **from Unit 1 to Unit 6** in **Computer Organization and Architecture (COA)**.

□ Part A: Multiple Choice (10 Questions)

1. Which type of memory is **fastest** to access?
   a) RAM
   b) Cache
   c) Hard Disk
   d) Magnetic Tape
2. In **K-map simplification**, what is the best group size to minimize expressions?
   a) 1
   b) 2
   c) Any power of 2
   d) Odd numbers only
3. Which of the following is **volatile** memory?
   a) ROM
   b) Hard Disk
   c) RAM
   d) SSD
4. A **multiplexer**:
   a) Demultiplexes signals
   b) Selects one of many inputs
   c) Performs arithmetic
   d) Stores data
5. A **half-adder** has:
   a) One input
   b) Two inputs and one output
   c) Two inputs and two outputs
   d) One input and two outputs
6. Which **RAID level** uses **mirroring**?
   a) RAID 0
   b) RAID 1
   c) RAID 5
   d) RAID 6
7. The **average access time** in memory hierarchy depends on:
   a) Cache size
   b) Hit ratio
   c) Main memory speed
   d) All of the above
8. **Micro-operations** are:
   a) High-level programming instructions
   b) Machine instructions

c) Elementary operations on data
d) Cache instructions

9. In a **two-level cache** system, **L1** cache is:
   a) Slower but larger
   b) Faster but smaller
   c) Same size as L2
   d) Shared among processors
10. A **cylinder** in a hard disk is:
   a) A single track
   b) A circular disk
   c) Set of aligned tracks across platters
   d) A magnetic region

□ Part B: True or False (10 Questions)

11. __ The control unit of the CPU manages the execution of instructions.
12. __ A NAND gate gives high output only when all inputs are high.
13. __ Optical storage like CD-ROM uses magnetic fields to read/write.
14. __ Micro-operations occur during the execution phase of an instruction cycle.
15. __ Magnetic tape allows random access to data.
16. __ The speed of a magnetic disk is affected by seek time and rotational latency.
17. __ RAID 0 provides redundancy by mirroring disks.
18. __ In a register transfer operation, data moves from one register to another.
19. __ A decoder converts binary information from n input lines to a maximum of 2^n output lines.
20. __ Larger cache always guarantees better performance.

✍ □ Part C: Fill in the Blanks (5 Questions)

21. A _____ operation transfers data from memory to a register.
22. The _____ performs arithmetic and logic operations in a CPU.
23. _____ is used to simplify Boolean algebra expressions using grouping techniques.
24. A _____ disk allows storage to be physically removed and inserted.
25. A control signal that triggers operations within the CPU is called a _____ signal.

✏ □ Part D: Short Answer (5 Questions)

26. What is the difference between **combinational** and **sequential** circuits?

27. Define **cache memory** and explain why it is needed.
28. Explain the difference between **seek time** and **latency** in a magnetic disk.
29. What is the function of the **control unit** in a CPU?
30. Briefly describe the purpose of **RAID** in storage systems.

Answer

Sure! Here's the **answer key** for all 30 questions based on COA Units 1 to 6:

□ Part A: Multiple Choice (Answers)

1. **b)** Cache
2. **c)** Any power of 2
3. **c)** RAM
4. **b)** Selects one of many inputs
5. **c)** Two inputs and two outputs
6. **b)** RAID 1
7. **d)** All of the above
8. **c)** Elementary operations on data
9. **b)** Faster but smaller
10. **c)** Set of aligned tracks across platters

□ Part B: True or False (Answers)

11. **True**
12. **False** (NAND gives LOW output only when all inputs are HIGH)
13. **False** (Optical storage uses laser, not magnetic field)
14. **True**
15. **False** (Magnetic tape is serial access, not random access)
16. **True**
17. **False** (RAID 1 provides redundancy; RAID 0 does not)
18. **True**
19. **True**
20. **False** (Larger cache doesn’t always mean better performance due to access time and hit/miss ratios)

✍ Part C: Fill in the Blanks (Answers)

21. **Read**
22. **ALU (Arithmetic Logic Unit)**
23. **Karnaugh Map (K-map)**
24. **Removable**
25. **Control**

✏ Part D: Short Answer (Sample Answers)

26. **Difference between combinational and sequential circuits:**

- *Combinational circuits* have outputs that depend only on current inputs.
- *Sequential circuits* depend on current inputs and past inputs (using memory elements like flip-flops).

27. **Cache memory:**

- A small, fast memory located close to the CPU.
- Stores frequently accessed data to speed up processing and reduce time spent accessing main memory.

28. **Seek time vs. Latency:**

- *Seek time:* Time taken to move the read/write head to the desired track.
- *Latency:* Time waiting for the desired sector to rotate under the head.

29. **Control Unit Function:**

- Directs the flow of data within the CPU.
- Fetches, decodes instructions and generates control signals for execution.

30. **Purpose of RAID:**

- RAID (Redundant Array of Independent Disks) is used to improve **data reliability, redundancy, and performance** in storage systems by distributing data and parity across multiple disks.